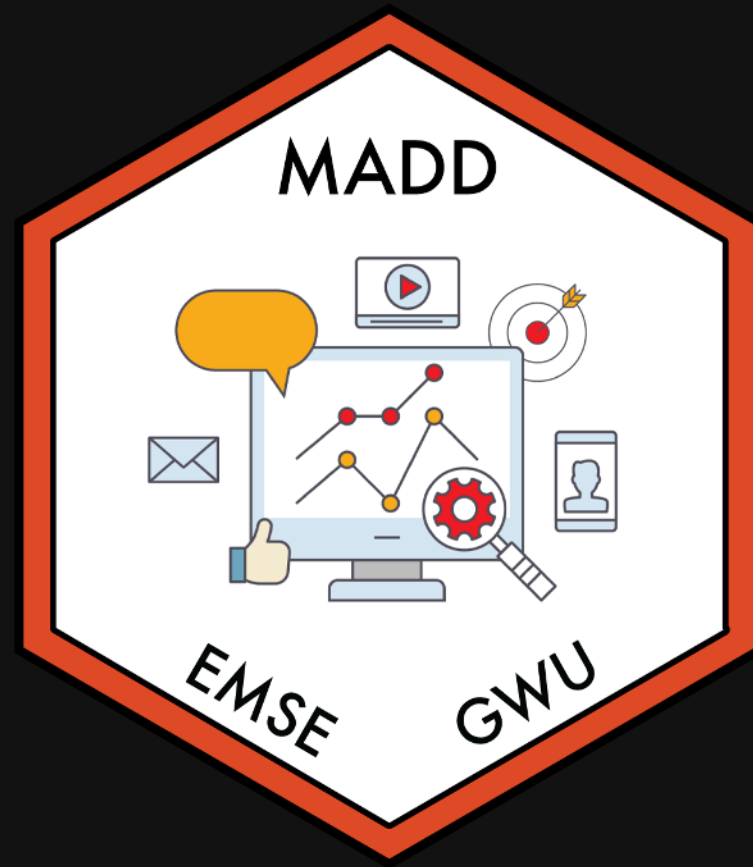




Conjoint Questions

🏛️ EMSE 6035: Marketing Analytics for Design Decisions

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Some Quarto tips

Convert a data frame to a markdown table with `kable()`

```
1 library(tidyverse)
2
3 mtcars %>%
4   kable()
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	1
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	1
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	2
Valiant	18.1	6	225.0	105	2.76	3.460	20.22	1	0	3	1
Duster 360	14.3	8	360.0	245	3.21	3.570	15.84	0	0	3	4
Merc 240D	24.4	4	146.7	62	3.69	3.190	20.00	1	0	4	2
Merc 230	22.8	4	140.8	95	3.92	3.150	22.90	1	0	4	2
Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4

Example from last year

```
1 library(tidyverse)
2
3 df <- read_csv("competitors.csv")
4 df %>%
5   kable()
```

brand	volumelnOz	quantity	price	pricePerCup	biodegradability	opacity	logo
greatValue	9	50	\$2.98	\$0.06	FALSE	TRUE	FALSE
decorRack	9	50	\$5.99	\$0.12	FALSE	TRUE	FALSE
tigerChef	9	100	\$7.99	\$0.08	FALSE	TRUE	FALSE
smartly	9	80	\$2.79	\$0.03	FALSE	FALSE	FALSE
solo	9	50	\$4.04	\$0.08	FALSE	FALSE	FALSE
greatValue	9	100	\$3.76	\$0.04	FALSE	FALSE	FALSE
ecoProducts	9	1000	\$187.69	\$0.19	TRUE	FALSE	TRUE
worldCentric	9	2000	\$220.00	\$0.11	TRUE	FALSE	TRUE
hefty	18	50	\$3.98	\$0.08	FALSE	TRUE	FALSE

More `kable()` formatting options:
`{kableExtra}` package

References

Simple approach: Insert a footnote with `^[]`

This markdown...

```
The Eiffel Tower is 324 meters  
tall^[From the [Eiffel Tower wikipedia  
page](https://en.wikipedia.org/wiki  
/Eiffel_Tower)]
```

...renders as this

The Eiffel Tower is 324 meters tall¹

¹From the [Eiffel Tower wikipedia page](https://en.wikipedia.org/wiki/Eiffel_Tower)

References

Complex (but more complete) approach: Use bibtex

<https://quarto.org/docs/authoring/footnotes-and-citations.html>

You can insert citations with `[@citekey]`, and a “References” table will be automatically created.

Footnotes are perfectly fine for this class

Week 7: *Conjoint Questions*

1. Defining choice questions in R
2. Displaying choice questions in surveydown

Week 7: *Conjoint Questions*

1. Defining choice questions in R
2. Displaying choice questions in surveydown

Download & unzip:

- logitr-cars
- today's practice notes (2 survey templates)

Open the `logitr-cars` folder in Positron

We'll be using the `{cbcTools}` package today

Install with:

```
1 remotes::install_github("jhelvy/cbcTools")
```

Choice question components

1. Generate `profiles` for each attribute and level
2. Create a survey `design` data frame from `profiles`

Basic Design

Any combination of attributes can be shown in each choice question

Question 1

Option:	1	2	3
Price:	\$15	\$25	\$25
Fuel Economy:	25 (mpg)	25 (mpg)	25 (mpg)
Accel. Time:	6 (s)	7 (s)	8 (s)
Powertrain:	Electric	Gasoline	Electric

Question 2

Option:	1	2	3
Price:	\$20	\$25	\$25
Fuel Economy:	25 (mpg)	20 (mpg)	25 (mpg)
Accel. Time:	7 (s)	6 (s)	7 (s)
Powertrain:	Gasoline	Gasoline	Electric

Labeled Design

One attribute is used as the "label" - choice options are fixed according to the label

Question 1

Option:	Gasoline	Electric
Price:	\$20	\$25
Fuel Economy:	20 (mpg)	25 (mpg)
Accel. Time:	8 (s)	8 (s)

Question 2

Option:	Gasoline	Electric
Price:	\$20	\$25
Fuel Economy:	20 (mpg)	20 (mpg)
Accel. Time:	6 (s)	8 (s)

Design with a “None” option

A “none” option means they can choose an “other” option

Question 1

Option:	1	2	3	None
Price:	\$15	\$20	\$15	
Fuel Economy:	25 (mpg)	25 (mpg)	30 (mpg)	
Accel. Time:	8 (s)	8 (s)	7 (s)	
Powertrain:	1	1	1	

Question 2

Option:	1	2	3	None
Price:	\$15	\$15	\$25	
Fuel Economy:	25 (mpg)	30 (mpg)	30 (mpg)	
Accel. Time:	6 (s)	8 (s)	6 (s)	
Powertrain:	1	2	2	

Attribute-specific features

Some attributes may only be valid for certain levels of other attributes

Example: The driving range of an electric vehicle (EV) only applies to EVs and not gasoline-powered vehicles.

To implement this, you may need to add restrictions to your `profiles` with `cbc_restrict()`
(see `logitr-cars` code 1.2)

Warning: Restrictions can lead to imbalances

Use `balance_by` in `cbc_design()` to fix imbalances

(see `logitr-cars` code 1.2)

Your Turn

20:00

Open the “conjoint_buttons” template folder in Positron

With your team, discuss the specific choice question design for your project

- Regular or labeled?
- Include a “none” option (outside good) or not?
- Include restrictions?

Then edit the `make-choice-questions.R` file to design your choice questions.

Week 7: *Conjoint Questions*

1. Defining choice questions in R
2. Displaying choice questions in surveydown

Displaying choice questions in your survey

1. Read in the design file (`choice_questions.csv`)
2. Sample and store a random respondent ID
3. Filter the design for the respondentID
4. Define a function to create question options
5. Create the options for each choice question
6. Create each choice question (6 in total)

1. Read in the design file (`choice_questions.csv`)

```
1 design <- readr::read_csv("choice_questions.csv")
```

2. Sample and store a random respondent ID

Sample a random respondentID

```
1 respondentID <- sample(design$respID, 1)
```

Store the respondentID

```
1 sd_store_value(respondentID, "respID")
```

In your survey data, you will have a `respID` column

3. Filter the design for the respondentID

```
1 df <- design %>%
2   filter(respID == respondentID) %>%
3   mutate(image = paste0("images/", image))
```

Images are stored in "images" folder, so we paste on that path

Result looks like this:

```
1 head(df)
```

```
#> # A tibble: 6 × 9
#>   profileID respID   qID altID obsID type      price freshness image
#>   <dbl>   <dbl> <dbl> <dbl> <dbl> <chr>    <dbl>   <chr>    <chr>
#> 1      29     399     1     1  2389 Pink Lady    3.5 Excellent images/pinkl
#> 2      68     399     1     2  2389 Honeycrisp  4 Average images/honey
#> 3      17     399     1     3  2389 Gala      2.5 Excellent images/gala
#> 4      33     399     2     1  2390 Honeycrisp  4 Excellent images/honey
#> 5      69     399     2     2  2390 Pink Lady    4 Average images/pinkl
#> 6      45     399     2     3  2390 Red Delicious 1.5 Average images/redDe
```

4. Define a function to create question options

```
1 make_cbc_options <- function(df) {  
2   alt1 <- df |> filter(altID == 1)  
3   alt2 <- df |> filter(altID == 2)  
4   alt3 <- df |> filter(altID == 3)  
5  
6   options <- c("option_1", "option_2", "option_3")  
7  
8   names(options) <- c(  
9     glue(  
10      "  
11      **Option 1**<br>  
12      <img src='{alt1$image}' width=100><br>  
13      **Type**: {alt1$type}<br>  
14      **Price**: $ {alt1$price} / lb  
15      "  
16    ),  
17    glue(  
18      "  
19      **Option 2**<br>  
20      <img src='{alt2$image}' width=100><br>  
21      **Type**: {alt2$type}<br>  
22      **Price**: $ {alt2$price} / lb  
23      "  
24    ),  
25    glue(  
26      "  
27      **Option 3**<br>  
28      <img src='{alt3$image}' width=100><br>  
29      **Type**: {alt3$type}<br>  
30      **Price**: $ {alt3$price} / lb  
31      "  
32    )  
33  )  
34  return(options)  
35 }
```

4. Define a function to create question options

Make one-row data frames for each alternative.

For example, `alt1` looks like this:

```
#> Rows: 1
#> Columns: 9
#> $ profileID <dbl> 29
#> $ respID    <dbl> 399
#> $ qID       <dbl> 1
#> $ altID     <dbl> 1
#> $ obsID     <dbl> 2389
#> $ type      <chr> "Pink Lady"
#> $ price     <dbl> 3.5
#> $ freshness <chr> "Excellent"
#> $ image     <chr> "images/pinkLady.jpg"
```

```
1 make_cbc_options <- function(df) {
2   alt1 <- df |> filter(altID == 1)
3   alt2 <- df |> filter(altID == 2)
4   alt3 <- df |> filter(altID == 3)
5
6   options <- c("option_1", "option_2", "option_3")
7
8   names(options) <- c(
9     glue(
10      "
11      **Option 1**<br>
12      <img src='{alt1$image}' width=100><br>
13      **Type**: {alt1$type}<br>
14      **Price**: $ {alt1$price} / lb
15      "
16    ),
17    glue(
18      "
19      **Option 2**<br>
20      <img src='{alt2$image}' width=100><br>
21      **Type**: {alt2$type}<br>
22      **Price**: $ {alt2$price} / lb
23      "
24    ),
25    glue(
26      "
27      **Option 3**<br>
28      <img src='{alt3$image}' width=100><br>
29      **Type**: {alt3$type}<br>
30      **Price**: $ {alt3$price} / lb
31      "
32    )
33  )
34  return(options)
35 }
```

4. Define a function to create question options

```
1 alt1[c('image', 'type', 'price')]
```

```
#> # A tibble: 1 × 3  
#>   image          type      price  
#>   <chr>         <chr>    <dbl>  
#> 1 images/pinkLady.jpg Pink Lady    3.5
```

Renders as:

```
#> **Option 1**<br>  
#> <img src='images/pinkLady.jpg' width=100><br>  
#> **Type**: Pink Lady<br>  
#> **Price**: $ 3.5 / lb
```

```
1 make_cbc_options <- function(df) {  
2   alt1 <- df |> filter(altID == 1)  
3   alt2 <- df |> filter(altID == 2)  
4   alt3 <- df |> filter(altID == 3)  
5  
6   options <- c("option_1", "option_2", "option_3")  
7  
8   names(options) <- c(  
9     glue(  
10      "  
11      **Option 1**<br>  
12      <img src='{alt1$image}' width=100><br>  
13      **Type**: {alt1$type}<br>  
14      **Price**: $ {alt1$price} / lb  
15      "  
16    ),  
17    glue(  
18      "  
19      **Option 2**<br>  
20      <img src='{alt2$image}' width=100><br>  
21      **Type**: {alt2$type}<br>  
22      **Price**: $ {alt2$price} / lb  
23      "  
24    ),  
25    glue(  
26      "  
27      **Option 3**<br>  
28      <img src='{alt3$image}' width=100><br>  
29      **Type**: {alt3$type}<br>  
30      **Price**: $ {alt3$price} / lb  
31      "  
32    )  
33  )  
34  return(options)  
35 }
```

5. Create the options for each choice question

```
1 cbc1_options <- make_cbc_options(df |> filter(qID == 1))
```

Value stored in your data:

```
#> option_1
```

Label:

```
#> [1] " **Option 1**<br>\n  <img src=
```

Renders as:

****Option 1****



****Type****: Pink Lady

****Price****: \$ 3.5 / lb

5. Create the options for each choice question

Now make the options for each question

```
1 cbc1_options <- make_cbc_options(df |> filter(qID == 1))
2 cbc2_options <- make_cbc_options(df |> filter(qID == 2))
3 cbc3_options <- make_cbc_options(df |> filter(qID == 3))
4 cbc4_options <- make_cbc_options(df |> filter(qID == 4))
5 cbc5_options <- make_cbc_options(df |> filter(qID == 5))
6 cbc6_options <- make_cbc_options(df |> filter(qID == 6))
```

6. Create each choice question (6 in total)

Code:

```
1 sd_question(  
2     type = 'mc_buttons',  
3     id = 'cbc_q1',  
4     label = "(1 of 6) If these were your only c  
5     option = cbc1_options  
6 )
```

Renders as:

Option 1  Type: Red Delicious Price: \$ 1 / lb	Option 2  Type: Red Delicious Price: \$ 2 / lb	Option 3  Type: Fuji Price: \$ 2 / lb
--	--	---

TABLE layout: Show options in a table with `kable()`

```
1 library(dplyr)
2
3 alts <- df %>%
4   filter(qID == 1) %>%
5   mutate(
6     # Add $ sign to price and html code for image path
7     price = paste(scales::dollar(price), "/ lb"),
8     image = paste0('<img src="", image, '" width=100>')
9   ) %>%
10  # Make nicer attribute labels
11  select(
12    `Option:` = altID,
13    ` ` = image,
14    `Price:` = price,
15    `Type:` = type,
16    `Freshness:` = freshness
17  )
18
19 # Drop row names
20 row.names(alts) <- NULL
```

Option: Show options in a table with `kable()`

Display the *transpose*, `t(alts)`

```
1 kbl(t(alts), escape = FALSE) %>%  
2   kable_styling(  
3     bootstrap_options = c("striped", "hover", "condensed"),  
4     full_width = FALSE,  
5     position = "center"  
6   )
```

Option:	1	2	3
			
Price:	\$3.50 / lb	\$4.00 / lb	\$2.50 / lb
Type:	Pink Lady	Honeycrisp	Gala
Freshness:	Excellent	Average	Excellent

TABLE layout: Show options in a table with `kable()`

Your choice question can be defined in the `survey.qmd` file as it is just choosing one of three options:

If these were your only options, which would you choose?

Option 1 Option 2 Option 3

Option:	1	2	3
			
Price:	\$4.00 / lb	\$2.50 / lb	\$4.00 / lb
Type:	Fuji	Gala	Honeycrisp
Freshness:	Excellent	Excellent	Average

Next

Your Turn

20:00

With your team, choose a format to work with (buttons or table) for your choice question, then modify the example survey to match your project.

Versions:

Buttons layout: `conjoint_buttons`

Table layout: `conjoint_tables`